

Requerments

1. Project Objective

The project implements a database system for managing bank customers, accounts, payment cards, card transactions, merchants and merchant categories.

The operational database will provide the source data for a Data Warehouse used to analyze historical card transaction activity.

2. Scope

The system covers:

- Customers
- Bank accounts
- Payment cards
- Card payment transactions
- Merchants
- Merchant categories

Loans, mortgages, bank transfers, investments, cash withdrawals, fraud detection, credit risk and other banking products are outside the scope.

3. Functional Requirements

The system shall:

- store customer and address information,
- manage customer accounts,
- support accounts with one or multiple owners,
- assign cards to a cardholder and an account,
- store card information and card status,
- record card transactions with amount, currency, date, time, type and status,

- assign each transaction to the card used and the merchant where it was made,
- store merchant information and addresses,
- classify merchants by merchant category.

4. Business Rules

- A customer can own multiple accounts, and an account can have multiple owners.
- The customer-account relationship contains a Role.
- A customer can hold multiple cards; each card has one holder.
- An account can have multiple cards; each card is linked to one account.
- A card can be used for multiple transactions; each transaction uses one card.
- A merchant can receive multiple transactions; each transaction is made at one merchant.
- A merchant belongs to one merchant category; one category can contain multiple merchants.

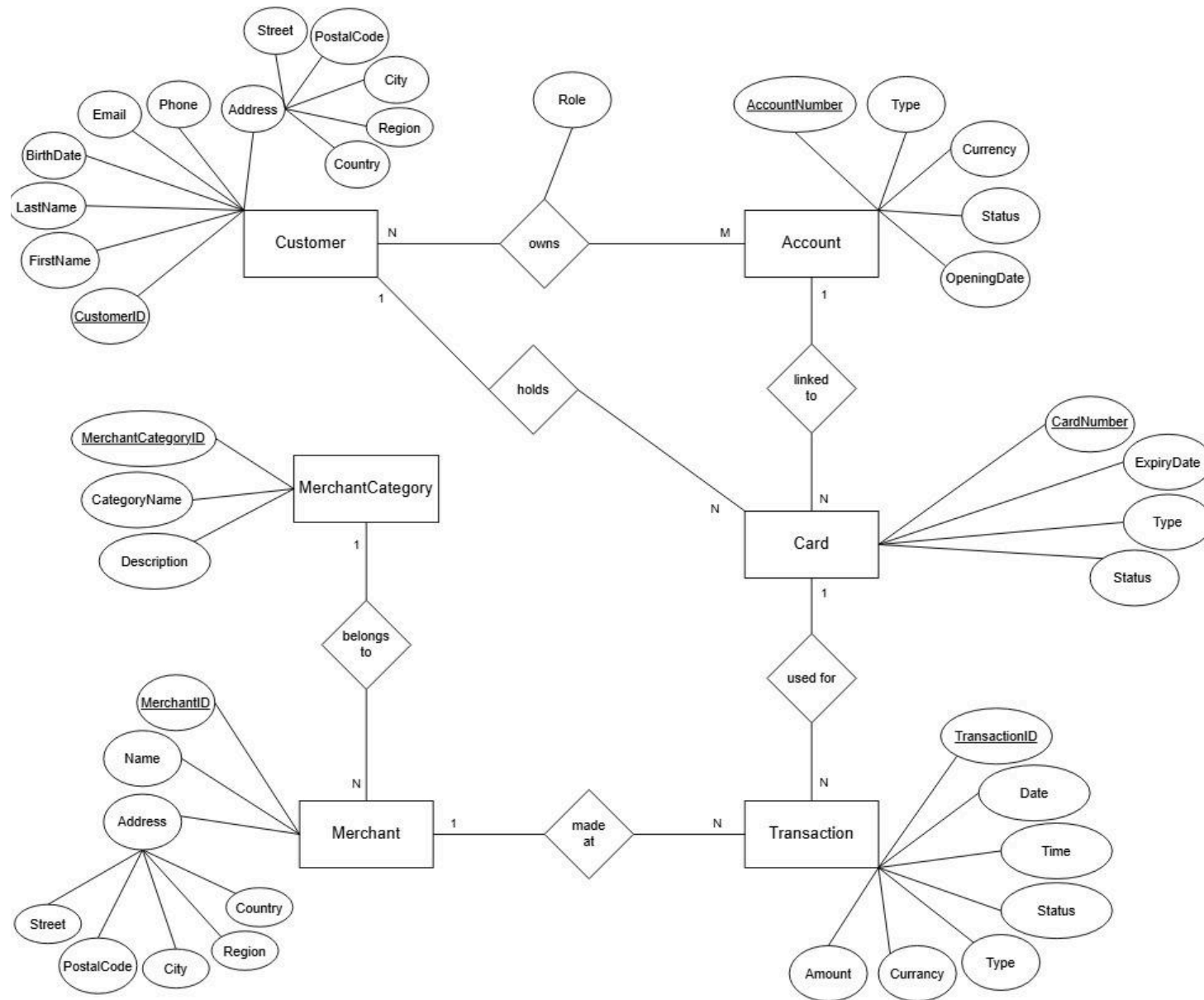
5. Database Design and Implementation

The database will be designed according to the following phases:

ER Model → Entity Characteristics → Relations → Dependency Diagrams → Normalization → Data Types → SQL Implementation

The resulting operational database will later serve as a source for the Data Warehouse and ETL process.

Entity-relationship model



Characteristics of Entities

Characteristic	Value
Entity	Customer
Estimated records	100,000
Estimated Growth	10,000/year
Primary Key	CustomerID
Attributes	CustomerID (integer), FirstName (nvarchar), LastName (nvarchar), BirthDate (date), Email (nvarchar), Phone (nvarchar), Street (nvarchar), PostalCode (nvarchar), City (nvarchar), Region (nvarchar), Country (nvarchar)

Characteristic	Value
Entity	Account
Estimated records	120,000
Estimated Growth	12,000/year
Primary Key	AccountNumber
Attributes	AccountNumber (nvarchar), Type (nvarchar), Currency (nvarchar), Status (nvarchar), OpeningDate (date)

Characteristic	Value
Entity	Card
Estimated records	150,000
Estimated Growth	20,000/year
Primary Key	CardNumber
Attributes	CardNumber (nvarchar), ExpiryDate (date), Type (nvarchar), Status (nvarchar)

Characteristic	Value
Entity	Transaction
Estimated records	5,000,000
Estimated Growth	2,000,000/year
Primary Key	TransactionID
Attributes	TransactionID (bigint), Date (date), Time (time), Status (nvarchar), Type (nvarchar), Currency (nvarchar), Amount (decimal)

Characteristic	Value
Entity	Merchant
Estimated records	50,000
Estimated Growth	5,000/year
Primary Key	MerchantID
Attributes	MerchantID (integer), Name (nvarchar), Street (nvarchar), PostalCode (nvarchar), City (nvarchar), Region (nvarchar), Country (nvarchar)

Characteristic	Value
Entity	MerchantCategory
Estimated records	100
Estimated Growth	-
Primary Key	MerchantCategoryID
Attributes	MerchantCategoryID (integer), CategoryName (nvarchar), Description (nvarchar)

Relational Model

Entities

Customer : {[CustomerID, FirstName, LastName, BirthDate, Email, Phone, Street, PostalCode, City, Region, Country]}

Account : {[AccountNumber, Type, Currency, Status, OpeningDate]}

Card : {[CardNumber, ExpiryDate, Type, Status]}

Transaction : {[TransactionID, Date, Time, Status, Type, Currency, Amount]}

Merchant : {[MerchantID, Name, Street, PostalCode, City, Region, Country]}

MerchantCategory : {[MerchantCategoryID, CategoryName, Description]}

Relations

Owns : {[CustomerID, AccountNumber, Role]}

Holds : {[CustomerID, CardNumber]}

LinkedTo : {[AccountNumber, CardNumber]}

UsedFor : {[CardNumber, TransactionID]}

MadeAt : {[MerchantID, TransactionID]}

BelongsTo : {[MerchantCategoryID, MerchantID]}

Simplification

Customer : {[CustomerID, FirstName, LastName, BirthDate, Email, Phone, Street, PostalCode, City, Region, Country]}

Account : {[AccountNumber, Type, Currency, Status, OpeningDate]}

Card : {[CardNumber, ExpiryDate, Type, Status, CustomerID, AccountNumber]}

Transaction : {[TransactionID, Date, Time, Status, Type, Currency, Amount, CardNumber, MerchantID]}

Merchant : {[MerchantID, Name, Street, PostalCode, City, Region, Country, MerchantCategoryID]}

MerchantCategory : {[MerchantCategoryID, CategoryName, Description]}

Owns : {[CustomerID, AccountNumber, Role]}

Holds : {[CardNumber, CustomerID]}

LinkedTo : {[CardNumber, AccountNumber]}

UsedFor : {[TransactionID, CardNumber]}

MadeAt : {[TransactionID, MerchantID]}

BelongsTo : {[MerchantID, MerchantCategoryID]}

Final Version

Customer : {[CustomerID, FirstName, LastName, BirthDate, Email, Phone, Street, PostalCode, City, Region, Country]}

Account : {[AccountNumber, Type, Currency, Status, OpeningDate]}

Card : {[CardNumber, ExpiryDate, Type, Status, **CustomerID**, **AccountNumber**]}

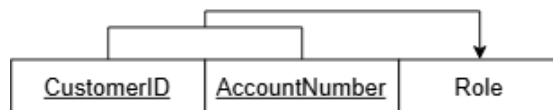
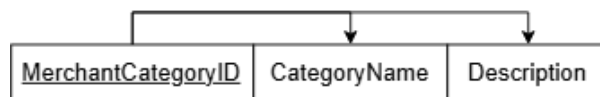
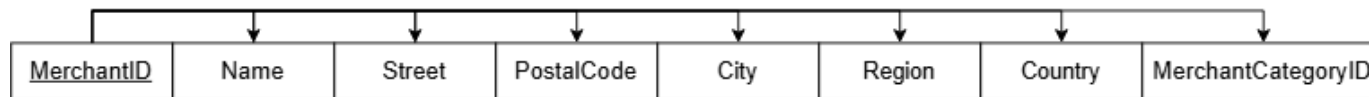
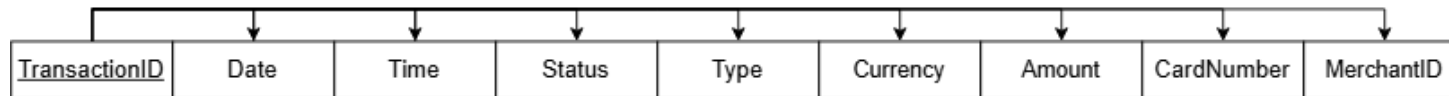
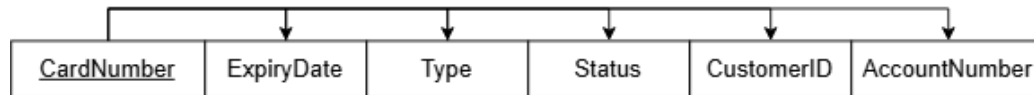
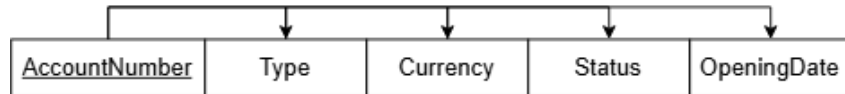
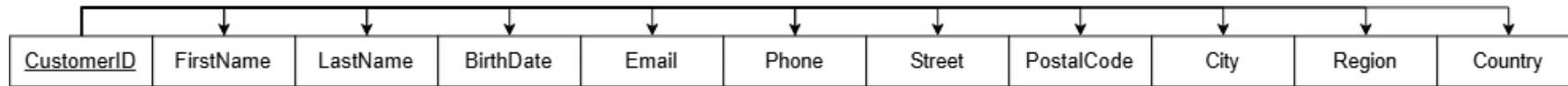
Transaction : {[TransactionID, Date, Time, Status, Type, Currency, Amount, **CardNumber**, **MerchantID**]}

Merchant : {[MerchantID, Name, Street, PostalCode, City, Region, Country, **MerchantCategoryID**]}

MerchantCategory : {[MerchantCategoryID, CategoryName, Description]}

Owns : {[**CustomerID**, **AccountNumber**, Role]}

Dependency Diagrams



Normalization Check

First Normal Form (1NF)

All relations are in 1NF because all attributes contain atomic values and there are no repeating groups.

Second Normal Form (2NF)

All relations are in 2NF. Most relations have a single-attribute primary key. In Owns, the attribute Role depends on the complete composite key (CustomerID, AccountNumber). Therefore, no partial dependencies exist.

Third Normal Form (3NF)

All relations are in 3NF because there are no transitive dependencies between non-key attributes.

Conclusion

All relations are already in Third Normal Form (3NF). No further decomposition is required.

Storage Calculation

Customer

Attribute	Data Type	Length	Storage / B	Description / Constraints	No. of Records	Total Storage / B
CustomerID	INT	–	4	PK, IDENTITY, NOT NULL	100,000	400,000
FirstName	NVARCHAR	50	100	NOT NULL	100,000	10,000,000
LastName	NVARCHAR	50	100	NOT NULL	100,000	10,000,000
BirthDate	DATE	–	3	NOT NULL	100,000	300,000
Email	NVARCHAR	100	200	Email address	100,000	20,000,000
Phone	NVARCHAR	25	50	Phone number	100,000	5,000,000
Street	NVARCHAR	100	200	NOT NULL	100,000	20,000,000
PostalCode	NVARCHAR	20	40	NOT NULL	100,000	4,000,000
City	NVARCHAR	100	200	NOT NULL	100,000	20,000,000
Region	NVARCHAR	100	200	Region / state	100,000	20,000,000
Country	NVARCHAR	60	120	NOT NULL	100,000	12,000,000
Total			1,217 B / record		100,000	121,700,000

Account

Attribute	Data Type	Length	Storage / B	Description / Constraints	No. of Records	Total Storage / B
AccountNumber	NVARCHAR	34	68	PK, NOT NULL	120,000	8,160,000
Type	NVARCHAR	20	40	NOT NULL	120,000	4,800,000
Currency	CHAR	3	3	ISO currency code, NOT NULL	120,000	360,000
Status	NVARCHAR	20	40	NOT NULL	120,000	4,800,000
OpeningDate	DATE	–	3	NOT NULL	120,000	360,000
Total			154 B / record		120,000	18,480,000

Card

Attribute	Data Type	Length	Storage / B	Description / Constraints	No. of Records	Total Storage / B
CardNumber	NVARCHAR	19	38	PK, NOT NULL	150,000	5,700,000
ExpiryDate	DATE	–	3	NOT NULL	150,000	450,000
Type	NVARCHAR	20	40	NOT NULL	150,000	6,000,000
Status	NVARCHAR	20	40	NOT NULL	150,000	6,000,000
CustomerID	INT	–	4	FK → Customer.CustomerID, NOT NULL	150,000	600,000
AccountNumber	NVARCHAR	34	68	FK → Account.AccountNumber, NOT NULL	150,000	10,200,000
Total			193 B / record		150,000	28,950,000

Transaction

Attribute	Data Type	Length	Storage / B	Description / Constraints	No. of Records	Total Storage / B
TransactionID	BIGINT	–	8	PK, IDENTITY, NOT NULL	5,000,000	40,000,000
Date	DATE	–	3	NOT NULL	5,000,000	15,000,000
Time	TIME	0	3	TIME(0), NOT NULL	5,000,000	15,000,000
Status	NVARCHAR	20	40	NOT NULL	5,000,000	200,000,000
Type	NVARCHAR	20	40	NOT NULL	5,000,000	200,000,000
Currency	CHAR	3	3	ISO currency code, NOT NULL	5,000,000	15,000,000
Amount	DECIMAL	12,2	9	Transaction amount, NOT NULL	5,000,000	45,000,000
CardNumber	NVARCHAR	19	38	FK → Card.CardNumber, NOT NULL	5,000,000	190,000,000
MerchantID	INT	–	4	FK → Merchant.MerchantID, NOT NULL	5,000,000	20,000,000
Total			148 B / record		5,000,000	740,000,000

Merchant

Attribute	Data Type	Length	Storage / B	Description / Constraints	No. of Records	Total Storage / B
MerchantID	INT	–	4	PK, IDENTITY, NOT NULL	50,000	200,000
Name	NVARCHAR	100	200	NOT NULL	50,000	10,000,000
Street	NVARCHAR	100	200	Merchant street	50,000	10,000,000
PostalCode	NVARCHAR	20	40	Postal code	50,000	2,000,000
City	NVARCHAR	100	200	NOT NULL	50,000	10,000,000
Region	NVARCHAR	100	200	Region / state	50,000	10,000,000
Country	NVARCHAR	60	120	NOT NULL	50,000	6,000,000
MerchantCategoryID	INT	–	4	FK → MerchantCategory.MerchantCategoryID, NOT NULL	50,000	200,000
Total			968 B / record		50,000	48,400,000

MerchantCategory

Attribute	Data Type	Length	Storage / B	Description / Constraints	No. of Records	Total Storage / B
MerchantCategoryID	INT	–	4	PK, IDENTITY, NOT NULL	100	400
CategoryName	NVARCHAR	50	100	NOT NULL, UNIQUE	100	10,000
Description	NVARCHAR	255	510	Description of category	100	51,000
Total			614 B / record		100	61,400

Owns

Attribute	Data Type	Length	Storage / B	Description / Constraints	No. of Records	Total Storage / B
CustomerID	INT	–	4	PK, FK → Customer.CustomerID, NOT NULL	132,000	528,000
AccountNumber	NVARCHAR	34	68	PK, FK → Account.AccountNumber, NOT NULL	132,000	8,976,000
Role	NVARCHAR	20	40	NOT NULL	132,000	5,280,000
Total			112 B / record		132,000	14,784,000

Overall storage

Calculation	Result
Total storage	972,375,400 B
Total storage / KB	949,585.35 KB
Total storage / MB	927.33 MB
Total storage / GB	0.906 GB